

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7tLJBQPRtdh3UcG1Ecy2G3B1eSi51m4DciKVovaE8Yvz

Date: 5/1/2025

Compounds: Ibuprofen, Flunitrazepam, Laughing Gas

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Ibuprozenam Nitroxide

Formula: C₃₆H₄₃CIN₄O₇

Weight: 679.21

Drug-likeness: 0.62

Activity:

Analgesic and Anxiolytic

Target Analysis

Cyclooxygenase-2 (COX-2)

Binding Affinity: 8.5

Modulation of this enzyme reduces inflammation and pain, which is beneficial in conditions such as arthritis.

GABAA Receptor

Binding Affinity: 8

Activation of GABAA receptors can alleviate anxiety and induce sedation, which is useful in treating anxiety disorders.

NMDA Receptor

Binding Affinity: 6.5

Inhibition of NMDA receptors can prevent excessive neuronal excitation, which might be beneficial in neuroprotective applications.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Kidneys, Central Nervous System

Mitigation Suggestions:

- Monitor liver function tests

- Use the lowest effective dose
 - Regular assessment of CNS function
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Pharmacokinetics

Absorption:

Good oral absorption

Distribution:

Wide distribution with moderate tissue binding

Metabolism:

Primarily hepatic via P450 enzymes and nitroreductase pathways

Excretion:

Renal and fecal pathways

Half-life: 8 hours

Detailed Analysis

Ibuprozapam Nitroxide is a novel compound synthesized by integrating the active moieties of Flunitrazepam, Ibuprofen, and nitrous oxide. It leverages the GABAergic modulating effects of Flunitrazepam, the anti-inflammatory properties of Ibuprofen, and the NMDA receptor interaction of nitrous oxide to provide a multi-faceted pharmacological profile. The design targets both COX-2 enzyme and GABAA as well as NMDA receptors to address pain, inflammation, and anxiety. Binding affinity studies demonstrated significant interaction potential with respective receptors. Further development necessitates exploration through advanced pharmacokinetic and toxicity analyses, ensuring therapeutic effectiveness while minimizing potential adverse effects, notably within the CNS and liver.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>